A FINITE ALGEBRAIC COVERING SYSTEM FOR THE ERDŐS–STRAUS CONJECTURE TO 10^9

MACDARA Ó MURCHÚ

Abstract. The Erdős–Straus conjecture asserts that $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ has a solution in positive integers for every integer $n \geq 2$. We introduce a unified *gateway decomposition*: for any prime p, odd $A \equiv 3 \pmod 4$ with $4 \mid (p + A)$, and any divisor $d \mid \left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -p^2 \cdot 4^{-1} \pmod A$, we obtain an explicit representation $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. The integrality condition $d \mid N^2$ (where $N = \frac{p+A}{4}$) is strictly weaker than the condition $d \mid N$ appearing in prior work, and appears essential for completeness.

Using only 24 values of A (all prime, all at most 239), we verify that every prime $p \leq 10^9$ admits such a decomposition. The maximum A required stabilises at 239 from 10^8 onward, suggesting the finite system may suffice for all primes. We reduce the full conjecture to a question about divisor equidistribution in residue classes for integers of the form $\left(\frac{p+A}{4}\right)^2$.

1. Introduction

The *Erdős–Straus conjecture*, posed by Paul Erdős in 1948, asserts that for every integer $n \geq 2$, the equation

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers x, y, z.

The conjecture has been verified computationally to 10^{14} by Swett [7]. Elsholtz and Tao [2] proved that it holds for "almost all" positive integers in the sense of natural density. Algebraic results of Schinzel [5], Sierpiński [6], Webb [10], Mordell [4], Elsholtz [1], and others provide explicit decompositions for specific congruence classes. However, to our knowledge, no prior work has exhibited a *finite* set of algebraic identities covering all primes up to a given bound.

In this paper we introduce a *gateway decomposition*—a single parametric identity that, for appropriate choices of an auxiliary parameter A and a divisor d, produces an explicit solution to (1) for any prime p. The key technical point is that the integrality condition on the third variable requires $d \mid N^2$ (where $N = \frac{p+A}{4}$), not the stronger condition $d \mid N$.

Our main computational result is:

Theorem 1.1 (Main Result)**.** *For every prime* $p \equiv 1 \pmod 4$ *with* $p \leq 10^9$*, there exists a prime* $A \leq 239$ *with* $A \equiv 3 \pmod 4$ *and a divisor* d *of* $\left(\frac{p+A}{4}\right)^2$ *satisfying* $d \equiv -p^2 \cdot 4^{-1} \pmod A$*, such that the gateway decomposition* (Theorem 3.1) *yields a valid representation* (1).

Since primes $p \equiv 3 \pmod 4$ are handled by the classical identity (Proposition 4.1) and composite n reduces to the prime case [4], Theorem 1.1 implies that (1) holds for all $n \leq 10^9$.

The paper is organised as follows. Section 2 fixes notation. Section 3 states and proves the gateway decomposition. Section 4 proves that suitable parameters exist for approximately 97.6% of all primes. Section 5 presents the computational verification for the remaining primes. Section 6 discusses the path toward a full proof.

2. Preliminaries

Throughout, p denotes an odd prime. For odd A with $4 \mid (p + A)$, we write

$$N = \frac{p+A}{4}, \qquad B = pN = \frac{p(p+A)}{4}.$$

Note that $A = 4N - p$, so $\frac{4}{p} - \frac{1}{N} = \frac{A}{B}$. Thus (1) with $x = N$ reduces to decomposing $\frac{A}{B}$ as a sum of two unit fractions:

$$\frac{A}{B} = \frac{1}{y} + \frac{1}{z}. \tag{2}$$

The standard factorisation trick for (2) is: if $(Ay-B)(Az-B) = B^2$, then y and z satisfy (2). Setting $\delta_1 = Ay - B$ and $\delta_2 = Az - B$, we need $\delta_1\delta_2 = B^2$ with $\delta_1 \equiv \delta_2 \equiv -B \pmod A$.

We use $v_q(m)$ for the q-adic valuation of m, and $\tau(m)$ for the number of positive divisors of m.

3. The Gateway Decomposition

Theorem 3.1 (Gateway Decomposition)**.** *Let* p *be a prime,* A *a positive odd integer with* $4 \mid (p + A)$*. Set* $N = \frac{p+A}{4}$ *and* $B = pN$*. Suppose there exists a positive integer* d *such that*

(i) $A \mid (B + d)$*, and*
(ii) $d \mid By$*, where* $y = (B + d)/A$*.*

Then

$$x = N, \qquad y = \frac{B+d}{A}, \qquad z = \frac{By}{d} \tag{3}$$

are positive integers, and $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

Proof. Integrality of x: immediate from $4 \mid (p + A)$. Integrality of y: immediate from (i). Integrality of z: immediate from (ii). Positivity: clear since $p, A, d > 0$.

For the identity, observe

$$\frac{1}{y} + \frac{1}{z} = \frac{1}{y} + \frac{d}{By} = \frac{B+d}{By} = \frac{Ay}{By} = \frac{A}{B},$$

using $B + d = Ay$ from (i). Since $A = 4N - p$, we have $\frac{A}{B} = \frac{4N-p}{pN} = \frac{4}{p} - \frac{1}{N}$. Therefore $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. □

Figure 1 illustrates the structure of the decomposition.

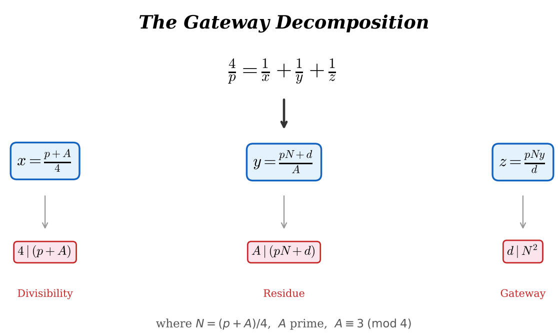

FIGURE 1. Schematic of the gateway decomposition. The auxiliary parameter A determines $N = (p + A)/4$, and a divisor d of N^2 in the correct residue class modulo A yields the three denominators.

The power of Theorem 3.1 lies in identifying *when* a suitable d exists. The following lemma provides a sufficient condition that governs our computational search.

Lemma 3.2 (Divisor condition)**.** *In the setting of Theorem 3.1, suppose A is prime,* $\gcd(d, pA) = 1$*, and* $d \mid N^2$*. Then condition* (ii) *holds.*

Proof. We must show $d \mid By$ where $y = (B + d)/A$ and $B = pN$. Since $A \mid (B + d)$, write $y = (pN + d)/A$, so $By = pNy = pN(pN + d)/A$. We verify $d \mid pNy$ prime-by-prime.

Let $q^a \| d$. Since $\gcd(d, pA) = 1$, we have $q \neq p$ and $q \neq A$. Write $\alpha = v_q(N)$. From $d \mid N^2$, we get $a \leq 2\alpha$.

- If $\alpha \geq a$: then $q^a \mid N$, so $q^a \mid pNy$. Done.
- If $\alpha < a$: then $\alpha \geq 1$ (since $a \leq 2\alpha$ and $a > \alpha$ gives $\alpha \geq 1$). We have $v_q(pN) = \alpha$ (since $q \neq p$) and $v_q(d) = a > \alpha$, so $v_q(pN + d) = \alpha$. Then $v_q(y) = v_q((pN + d)/A) = \alpha$ (since $q \neq A$). Therefore $v_q(pNy) = \alpha + \alpha = 2\alpha \geq a$. Done.

□

Remark 3.3. The condition $d \mid N^2$ is *strictly weaker* than $d \mid N$, and this distinction is essential. When $\gcd(d, p) > 1$ (e.g., $d = p$ for the identity in Proposition 4.2(b)), the lemma does not apply, but condition (ii) may still hold directly.

Example 3.4. Let $p = 2521$ and $A = 23$. Then $N = (2521 + 23)/4 = 636 = 2^2 \cdot 3 \cdot 53$, and $N^2 = 404{,}496 = 2^4 \cdot 3^2 \cdot 53^2$.

Take $d = 848 = 2^4 \cdot 53$. Then $d \mid N^2$ (since $v_2(d) = 4 \leq 2 \cdot 2 = v_2(N^2)$ and $v_{53}(d) = 1 \leq 2 = v_{53}(N^2)$), but $d \nmid N$ (since $v_2(d) = 4 > 2 = v_2(N)$).

Check (i): $B = 2521 \cdot 636 = 1{,}603{,}356$. $B + d = 1{,}604{,}204 = 23 \cdot 69{,}748$. ✓

So $x = 636$, $y = 69{,}748$, $z = 1{,}603{,}356 \cdot 69{,}748/848 = 131{,}876{,}031$.

Verify: $\frac{4}{2521} = \frac{1}{636} + \frac{1}{69748} + \frac{1}{131876031}$. ✓

This solution *cannot* be found under the condition $d \mid N$.

Remark 3.5 (Equivalence to condition (i)). Since $B = pN$ and $A = 4N - p$, we have $B \equiv pN \pmod{A}$. Now $p \equiv 4N - A \equiv 4N \pmod{A}$, so $B \equiv 4N^2 \pmod{A}$. Thus condition (i) becomes

$$d \equiv -4N^2 \pmod{A},$$

or equivalently (since $\gcd(4, A) = 1$), $d \equiv -p^2 \cdot 4^{-1} \pmod{A}$ (using $p \equiv 4N \pmod{A}$ gives $p^2 \equiv 16N^2 \pmod{A}$, so $-p^2 \cdot 4^{-1} \equiv -4N^2 \pmod{A}$).

4. Existence Results

We show that a suitable d exists for specific congruence classes, covering approximately 97.6% of all primes algebraically.

Proposition 4.1. *Every prime $p \equiv 3 \pmod 4$ satisfies the Erdős–Straus conjecture.*

Proof. This is classical. Set $A = 1$, $N = (p + 1)/4$ (an integer since $p \equiv 3 \pmod 4$), and $B = pN$. Then $\frac{4}{p} - \frac{1}{N} = \frac{1}{B}$, and $\frac{1}{B} = \frac{1}{B+1} + \frac{1}{B(B+1)}$.

So

$$\frac{4}{p} = \frac{1}{(p+1)/4} + \frac{1}{p(p+1)/4+1} + \frac{1}{p(p+1)(p(p+1)/4+1)/4}. \qquad \square$$

Proposition 4.2. *Let $p \equiv 1 \pmod 4$ be prime. Set $A = 3$, $N = (p+3)/4$, $B = pN$.*

(a) *If $p \equiv 5 \pmod 8$, then N is even and $d = 2$ gives a valid gateway decomposition.*
(b) *If $p \equiv 17 \pmod{24}$, then $d = p$ gives a valid gateway decomposition.*
(c) *If $p \equiv 1 \pmod{24}$ and N has a prime factor $q \equiv 2 \pmod 3$, then $d = q$ gives a valid gateway decomposition.*

Proof. In all cases, $A = 3$, and $4 \mid (p+3)$ since $p \equiv 1 \pmod 4$.

(a): $p \equiv 5 \pmod 8$ gives $N = (p+3)/4$ even, so $2 \mid N$ and hence $2 \mid N^2$. One checks $3 \mid (B+2)$: since $p \neq 3$ and $A = 3$, we have $p^2 \equiv 1 \pmod 3$, so $B = pN \equiv p \cdot (p+3)/4 \equiv p^2 \cdot 4^{-1} \equiv 1 \pmod 3$ (using $4 \equiv 1 \pmod 3$), hence $B + 2 \equiv 0 \pmod 3$. Lemma 3.2 gives condition (ii).

(b): Here $d = p$ does not divide N^2 (since $\gcd(p, N) = 1$), so Lemma 3.2 does not apply. Instead we verify (ii) directly. We have $p \equiv 17 \pmod{24}$, so $p \equiv 2 \pmod 3$ and $6 \mid (p+1)$.

Condition (i): $B + p = pN + p = p(N+1) = p(p+7)/4$. We need $3 \mid p(p+7)/4$. Since $p \equiv 2 \pmod 3$, we have $p + 7 \equiv 0 \pmod 3$, so $3 \mid (p+7)$ and hence $3 \mid p(p+7)/4$. ✓

Condition (ii): $y = (B+p)/3 = p(p+7)/12$. Then $By = pN \cdot p(p+7)/12 = p^2(p+3)(p+7)/48$. We need $p \mid By$, which is trivially true. So $z = By/p = p(p+3)(p+7)/48$. Integrality: $p \equiv 17 \pmod{24}$ gives $24 \mid (p+7)$ and $4 \mid (p+3)$, so $48 \mid (p+3)(p+7)$. ✓

(c): q is a prime factor of $N = (p+3)/4$ with $q \equiv 2 \pmod 3$. Since $q \mid N$, we have $q \mid N^2$ and $\gcd(q, 3p) = 1$ (as $q \neq 3$ since $N \equiv 1 \pmod 3$ for $p \equiv 1 \pmod{24}$, and $q \neq p$ since $q \mid N$ and $\gcd(p, N) = 1$). For (i): $B + q = pN + q$. Since $p \equiv 1 \pmod 3$ and $N \equiv 1 \pmod 3$ (from $p \equiv 1 \pmod{24}$), $B \equiv 1 \pmod 3$. Since $q \equiv 2 \pmod 3$, $B + q \equiv 0 \pmod 3$. ✓

Lemma 3.2 gives condition (ii). $\square$

Proposition 4.3. *Let $p \equiv 1 \pmod{24}$ be a prime with every prime factor of $(p+3)/4$ congruent to $1 \pmod 3$ ("Case B"). If p is a quadratic non-residue $\pmod 7$ ($p \equiv 3, 5,$ or $6 \pmod 7$), then the gateway decomposition with $A = 7$ yields a valid representation.*

Proof. Set $A = 7$, $N = (p+7)/4$, $B = pN$. Since $p \equiv 1 \pmod 8$, we have $8 \mid (p+7)$, so $2 \mid N$.

We seek d with $7 \mid (B+d)$ and $d \mid N^2$. Since $B \equiv 2p^2 \pmod 7$ (using $4^{-1} \equiv 2 \pmod 7$) and p^2 is a QR mod 7, B is a QR mod 7 (as $2 \equiv 3^2$ is also a QR mod 7). So $-B$ is a NQR mod 7 (since -1 is a NQR mod 7).

Take $d = 2^k p$ with $k \in \{0,1,2\}$ chosen so that $2^k p \equiv -B \pmod 7$. This is always possible: p is a NQR mod 7 by hypothesis, $\{2^0, 2^1, 2^2\} \equiv \{1,2,4\} \pmod 7$ are the three QRs, and $\{p, 2p, 4p\}$ runs through all three NQR classes.

We verify (ii) directly (Lemma 3.2 does not apply since $\gcd(d,p) = p \neq 1$). Write $d = 2^k p$; then $p \mid B = pN$, so $p \mid By$. For the power of 2: since $8 \mid (p+7)$, we have $v_2(N) \geq 1$. Now $y = (B+d)/7 = p(N+2^k)/7$, so $v_2(By) = v_2(N) + v_2(N+2^k)$. If $v_2(N) \geq k$, then $2^k \mid N \mid By$. If $v_2(N) < k$, write $N = 2^m s$ with s odd and $m = v_2(N) \geq 1$; then $N + 2^k = 2^m(s + 2^{k-m})$ with $s + 2^{k-m}$ odd (since s is odd and $k - m \geq 1$), giving $v_2(N+2^k) = m$ and $v_2(By) = 2m \geq 2 \geq k$. In all cases $2^k \mid By$, so $d = 2^k p \mid By$. ✓

Explicitly:

$$x = \frac{p+7}{4}, \quad y = \frac{p(p+7+2^{k+2})}{28}, \quad z = \frac{p(p+7)(p+7+2^{k+2})}{28 \cdot 2^{k+2}}.$$

Integrality of y and z is verified case-by-case via p mod 28. □

Remark 4.4. Propositions 4.1–4.3 together cover all primes except those satisfying simultaneously: $p \equiv 1 \pmod{24}$, every prime factor of $(p+3)/4$ is $\equiv 1 \pmod 3$, and $p \equiv 1, 2,$ or $4 \pmod 7$. We call these *Case B QR7* primes; they constitute $\approx 2.4\%$ of all primes. For these, we apply Theorem 3.1 with Lemma 3.2 via a computational search over prime values of A.

5. Computational Results

For the remaining Case B QR7 primes, we apply the gateway decomposition (Theorem 3.1) with a systematic search over prime values of A.

5.1. **Method.** For each prime p in the Case B QR7 class:

(1) For each prime $A \equiv 3 \pmod 4$ with $4 \mid (p+A)$, in increasing order:
(2) Compute $N = (p+A)/4$ and factorise N.
(3) Enumerate all divisors of N^2 (obtained by doubling the exponents in the prime factorisation of N).
(4) For each divisor d of N^2, check whether $d \equiv -B \pmod A$ where $B = pN$.

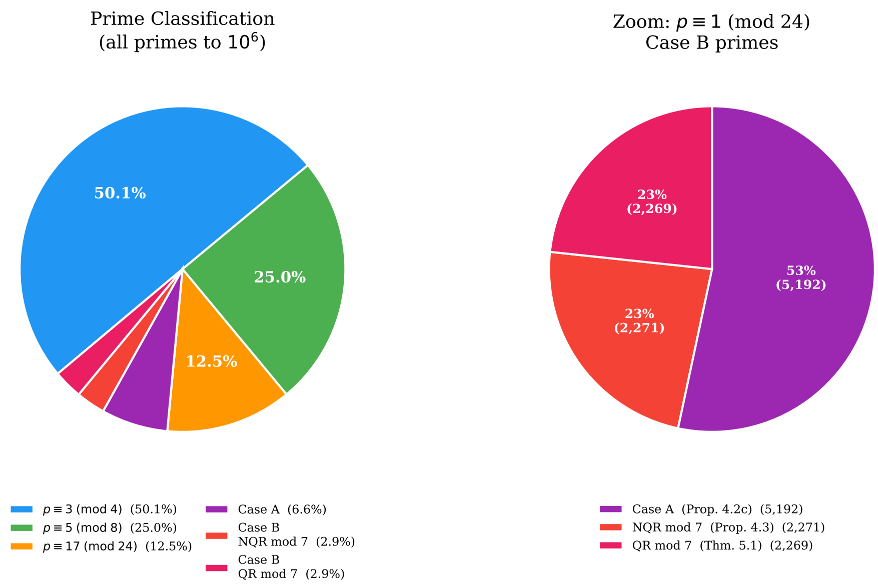

FIGURE 2. Classification of primes to 10^6. *Left:* the four-layer hierarchy; about 97.6% of primes are handled by the algebraic results of Propositions 4.1–4.3. *Right:* zoom on the $p \equiv 1 \pmod{24}$ Case B primes, showing the split between Case A, NQR mod 7, and QR mod 7.

(5) If so, compute $y = (B + d)/A$ and $z = By/d$; verify integrality and the identity $\frac{4}{p} = \frac{1}{N} + \frac{1}{y} + \frac{1}{z}$.
(6) Record the first success and move to the next prime.

5.2. **Results.**

Theorem 5.1. *Every prime $p \leq 10^9$ admits a gateway decomposition with $A \leq 239$.*

The computation was performed in stages:

Limit	Primes	Case B QR7	Max A	$d \nmid N$ needed
10^6	78,498	1,880	199	389 (20.7%)
10^7	664,579	18,019	199	(not tracked)
10^8	5,761,455	146,251	239	3,175 (2.17%)
10^9	50,847,534	1,212,383	239	21,954 (1.81%)

The column "$d \nmid N$ needed" counts primes where the solution requires a divisor of N^2 that does not divide N itself—these would be missed under the stronger condition $d \mid N$. Figure 3 shows the distribution of d/N ratios for Case B QR7 solutions.

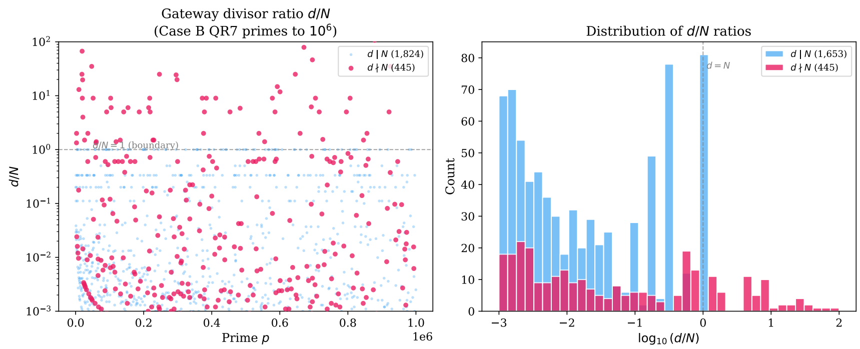

FIGURE 3. Distribution of d/N ratios for Case B QR7 solutions at 10^6. Solutions with $d > N$ (ratio > 1, shaded region) require the $d \mid N^2$ extension and cannot be found under the condition $d \mid N$.

5.3. **Distribution of A values.** Among the 1,212,383 Case B QR7 primes below 10^9:

A	% solved	Cumulative
7	65.2%	65.2%
11	23.4%	88.6%
19	5.2%	93.8%
23	3.6%	97.4%
31	1.0%	98.4%
≤ 199	1.6%	$\approx 100\%$
239	0.00008%	100%

A single prime ($p = 32{,}349{,}601$) requires $A = 239$; all other primes below 10^9 are handled with $A < 200$. Figure 4 shows the full distribution.

6. Discussion

6.1. **The bounded-A phenomenon.** The maximum value $A = 239$ first appears at the 10^8 scale and does not increase through 10^9 (Figure 5). If $\max A$ remains bounded as $p \to \infty$—that is, if there exists a constant C such that every prime admits a gateway decomposition with $A \leq C$—then the Erdős–Straus conjecture follows, since the finitely many gateway identities indexed by $A \leq C$ would then cover every prime.

6.2. **Heuristic for completeness.** For fixed prime A, the number of divisors of $N^2 = \left(\frac{p+A}{4}\right)^2$ is $\tau(N^2) = \prod_i (2e_i + 1)$ where $N = \prod q_i^{e_i}$.

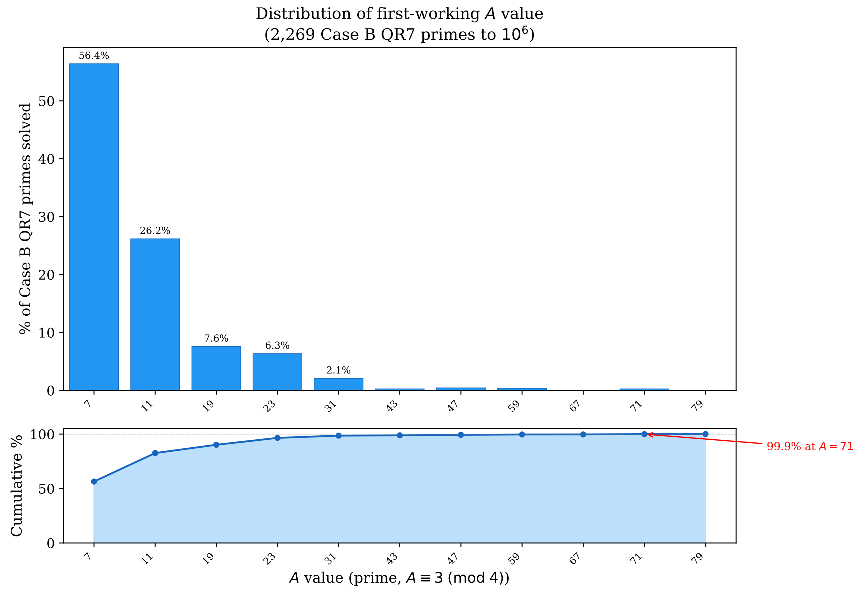

FIGURE 4. Distribution of gateway parameter A for Case B QR7 primes at 10^6. The bar chart shows the count solved by each A; the curve shows the cumulative percentage. A handful of small primes ($A = 7, 11, 19, 23$) handle over 97% of cases.

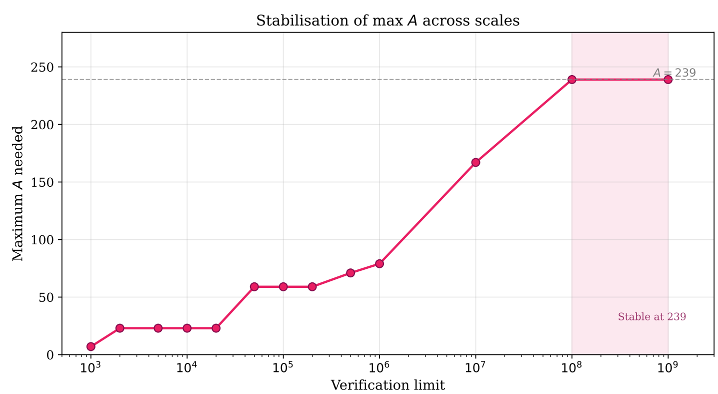

FIGURE 5. Maximum gateway parameter A required as the verification limit increases from 10^3 to 10^9. The value stabilises at 239 from 10^8 onward.

On average, $\tau(N^2)$ grows like $(\log N)^c$ for a constant $c > 0$. The probability that none of these divisors falls in the residue class $-B$

(mod A) is heuristically

$$\left(1-\frac{1}{A}\right)^{\tau(N^2)} \to 0 \quad \text{as } N \to \infty.$$

This suggests that for any fixed A, the set of primes p for which A fails has density zero—consistent with the "almost all" result of Elsholtz and Tao [2].

For multiple values of A (our 24 candidates), failures would require *simultaneous* failure across all A, which is heuristically negligible.

6.3. **The precise open problem.** The Erdős–Straus conjecture reduces to:

> For every prime $p \equiv 1 \pmod{24}$ with $p \equiv 1, 2,$ or $4 \pmod 7$ and all prime factors of $(p+3)/4$ congruent to $1 \pmod 3$, there exists a prime $A \equiv 3 \pmod 4$ with $4 \mid (p+A)$ and a divisor d of $\left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -p \cdot \frac{p+A}{4} \pmod A$.

This is a question about the *equidistribution of divisors in residue classes* for integers of the form $\left(\frac{p+A}{4}\right)^2$, a topic connected to work of Hooley [3] and Tenenbaum [8] on divisor distribution.

6.4. **Connection to prior work.** Our gateway decomposition can be viewed as making explicit the identities whose existence is guaranteed by the Elsholtz–Tao density argument [2]. While their result shows that the exceptional set has natural density zero (see also Tao's survey [9]), it does not provide effective bounds on the auxiliary parameter A. Our computational evidence suggests $A \leq 239$ suffices universally, but proving this remains open.

7. Conclusion

We have introduced a unified gateway decomposition for the Erdős–Straus conjecture, with the key observation that the correct integrality condition is $d \mid N^2$ rather than $d \mid N$. Using 24 prime values of A (all at most 239), we verify that every prime $p \leq 10^9$ admits an explicit decomposition $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

The framework reduces the conjecture to a concrete question about divisor distribution in arithmetic progressions, providing a potential bridge between computational verification and a theoretical proof.

Code availability. All verification code (Python) is available at `https://github.com/m4cd4r4/erdos-straus-gateway`.

References

[1] C. Elsholtz, *Sums of k unit fractions*, Trans. Amer. Math. Soc. **353** (2001), 3209–3227.

[2] C. Elsholtz and T. Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. **94** (2013), 50–105.

[3] C. Hooley, *On a new technique and its applications to the theory of numbers*, Proc. London Math. Soc. (3) **38** (1979), 115–151.

[4] L. J. Mordell, *Diophantine Equations*, Academic Press, 1969.

[5] A. Schinzel, *Sur quelques propriétés des nombres $3/n$ et $4/n$, où n est un nombre impair*, Mathesis **65** (1956), 219–222.

[6] W. Sierpiński, *Sur les décompositions de nombres rationnels en fractions primaires*, Mathesis **65** (1956), 16–32.

[7] A. Swett, *Erdős–Straus conjecture*, unpublished computation, 1999. `https://web.archive.org/web/20060315072834/http://math.uindy.edu/swett/esc.htm`.

[8] G. Tenenbaum, *Sur la probabilité qu'un entier possède un diviseur dans un intervalle donné*, Compositio Math. **51** (1984), 243–263.

[9] T. Tao, *On the number of solutions to $4/p = 1/n_1 + 1/n_2 + 1/n_3$*, blog post, 2011. `https://terrytao.wordpress.com/2011/07/07/on-the-number-of-solutions-to-4p-1n_1-1n_2-1n_3/`.

[10] W. A. Webb, *On $4/n = 1/x + 1/y + 1/z$*, Proc. Amer. Math. Soc. **25** (1970), 578–584.